```
import numpy as np
import matplotlib.pyplot as plt

# ==========================================
# STEP 1 & 4: DATA CENTER SIMULATION (FITNESS FUNCTION)
# ==========================================
def fitness_function(position):
    """
    Calculates the total power consumption of the data center.
    Position format: [Chiller_Temperature, AC_Fan_Speed]
    Goal: Minimize this value.
    """
    chiller_temp = position[0]
    fan_speed = position[1]

    # 1. Chiller Power: Lower setpoints require significantly more power
    p_chiller = 500.0 / (chiller_temp - 15.0 + 1e-5)

    # 2. Fan Power: Power scales cubically with fan speed
    p_fan = 0.5 * (fan_speed ** 3)

    # 3. Server Temperature Simulation
    # Higher chiller temp or lower fan speed makes servers hotter
    server_temp = 12.0 + (chiller_temp * 0.8) - (fan_speed * 0.15)

    # 4. Overheating Penalty
    # Severe penalty if servers exceed the safe operational limit (25°C)
    penalty = 0
    if server_temp > 25.0:
        penalty = 5000 + (server_temp - 25.0) * 1000

    total_power = p_chiller + p_fan + penalty
    return total_power

# ==========================================
# GREY WOLF OPTIMIZER IMPLEMENTATION
# ==========================================
def grey_wolf_optimizer(num_wolves=30, max_iter=100):
    # Problem boundaries
    # Dim 0: Chiller Temp (16°C to 24°C)
    # Dim 1: Fan Speed (10% to 100% capacity)
    lb = np.array([16.0, 10.0])
    ub = np.array([24.0, 100.0])
    dim = 2

    # STEP 3: Initialize Population
    Positions = np.random.uniform(low=lb, high=ub, size=(num_wolves, dim))

    # Initialize Alpha, Beta, and Delta positions and scores
    Alpha_pos = np.zeros(dim)
    Alpha_score = float("inf")

    Beta_pos = np.zeros(dim)
    Beta_score = float("inf")

    Delta_pos = np.zeros(dim)
    Delta_score = float("inf")

    # Track performance for convergence plot
    convergence_curve = []

    # Main Optimization Loop
    for l in range(max_iter):

        # STEP 4 & 5: Evaluate Fitness and Update Hierarchy
        for i in range(num_wolves):
            # Clip positions to keep them inside allowed data center operating ranges
            Positions[i] = np.clip(Positions[i], lb, ub)

            # Calculate power consumption (fitness)
            fitness = fitness_function(Positions[i])

            # Update Alpha (Best)
            if fitness < Alpha_score:
                Alpha_score = fitness
```

```
                Alpha_pos = Positions[i].copy()

            # Update Beta (2nd Best)
            elif fitness < Beta_score and fitness > Alpha_score:
                Beta_score = fitness
                Beta_pos = Positions[i].copy()

            # Update Delta (3rd Best)
            elif fitness < Delta_score and fitness > Beta_score and fitness > Alpha_score:
                Delta_score = fitness
                Delta_pos = Positions[i].copy()

        # STEP 7: Linearly decrease 'a' parameter from 2 to 0
        a = 2 - l * (2 / max_iter)

        # STEP 6: Update Positions of all Omega Wolves
        for i in range(num_wolves):
            for j in range(dim):

                # Equations matching Step 5 from the lab sheet:
                # Update against Alpha
                r1, r2 = np.random.rand(), np.random.rand()
                A1 = 2 * a * r1 - a
                C1 = 2 * r2
                D_alpha = abs(C1 * Alpha_pos[j] - Positions[i, j])
                X1 = Alpha_pos[j] - A1 * D_alpha

                # Update against Beta
                r1, r2 = np.random.rand(), np.random.rand()
                A2 = 2 * a * r1 - a
                C2 = 2 * r2
                D_beta = abs(C2 * Beta_pos[j] - Positions[i, j])
                X2 = Beta_pos[j] - A2 * D_beta

                # Update against Delta
                r1, r2 = np.random.rand(), np.random.rand()
                A3 = 2 * a * r1 - a
                C3 = 2 * r2
                D_delta = abs(C3 * Delta_pos[j] - Positions[i, j])
                X3 = Delta_pos[j] - A3 * D_delta

                # Take the average to determine new cooling setting
                Positions[i, j] = (X1 + X2 + X3) / 3

        convergence_curve.append(Alpha_score)

        # Print update every 10 iterations
        if (l + 1) % 10 == 0 or l == 0:
            print(f"Iteration {l+1}/{max_iter} | Current Min Power: {Alpha_score:.2f} kW")

    return Alpha_pos, Alpha_score, convergence_curve

# ==========================================
# EXECUTE AND PLOT RESULTS
# ==========================================
print("--- Starting Data Center Cooling Optimization via GWO ---")
best_settings, min_power, history = np.empty(2), 0, []
best_settings, min_power, history = grey_wolf_optimizer(num_wolves=30, max_iter=100)

print("\n==========================================")
print("🎯 OPTIMAL COOLING CONFIGURATION FOUND:")
print("==========================================")
print(f"🔹 Chiller Temperature Setpoint : {best_settings[0]:.2f} °C")
print(f"🔹 AC Fan Speed Capacity        : {best_settings[1]:.2f} %")
print(f"⚡ Minimum Total Power Draw     : {min_power:.2f} kW")
print("==========================================")

# Generate Convergence Plot (Matches the visual in Step 7 of your lab sheet)
plt.figure(figsize=(8, 5))
plt.plot(history, color='blue', linewidth=2)
plt.title('Convergence Plot: Data Center Cooling Optimization')
plt.xlabel('Iteration')
plt.ylabel('Best Objective Value (Power in kW)')
plt.grid(True)
plt.show()
```

```
--- Starting Data Center Cooling Optimization via GWO ---
Iteration 1/100 | Current Min Power: 6831.80 kW
Iteration 10/100 | Current Min Power: 662.09 kW
Iteration 20/100 | Current Min Power: 660.38 kW
Iteration 30/100 | Current Min Power: 660.38 kW
Iteration 40/100 | Current Min Power: 660.38 kW
Iteration 50/100 | Current Min Power: 660.38 kW
Iteration 60/100 | Current Min Power: 660.38 kW
Iteration 70/100 | Current Min Power: 660.05 kW
Iteration 80/100 | Current Min Power: 660.05 kW
Iteration 90/100 | Current Min Power: 660.05 kW
Iteration 100/100 | Current Min Power: 660.05 kW

========================================
🎯 OPTIMAL COOLING CONFIGURATION FOUND:
========================================
🔹 Chiller Temperature Setpoint : 18.12 °C
🔹 AC Fan Speed Capacity        : 10.00 %
⚡ Minimum Total Power Draw     : 660.05 kW
========================================
```

